ZAEEM MOHAMMED

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PROFESSIONAL SUMMARY

Mechatronics engineering student building an 18-DOF hexapod robotic vacuum cleaner for autonomous stair climbing (final year major project, team of 4), contributing the kinematics, physics simulation, Raspberry Pi control software and operator dashboard, with hands-on experience in embedded systems, sensor integration, and CAD/simulation tools (SolidWorks, Siemens NX, Robcad). Core interest in industrial automation and controls (PLC, SCADA), with a focused goal of building toward automation/mechatronics roles in smart-infrastructure and manufacturing environments.

TECHNICAL SKILLS

Automation & Controls:	PLC & SCADA, Ladder Logic, HDL
Embedded Systems:	ESP32, Arduino, Raspberry Pi, Sensor Integration (IMU, ToF, Gas/Temp)
Programming:	C, C++, Python, MATLAB, Assembly, NumPy, Matplotlib
Mechanical Design:	SolidWorks, AutoCAD, Fusion 360, Siemens NX
Simulation Tools:	PyBullet, MATLAB/Simulink, Ansys, Siemens Robcad

FEATURED PROJECTS

Self Stair-Climbing Hexapod Robotic Vacuum Cleaner

Final Year Project (Team of 4)

Technologies: Python, PyBullet, NumPy, Matplotlib, Inverse Kinematics, Raspberry Pi 4, PCA9685, BNO055 IMU, VL53L0X ToF, 3D Printing

- Developed with a team of 4 a spider-inspired 18-DOF hexapod robotic vacuum cleaner designed to climb stairs autonomously and carry a 2 kg payload, built to a total hardware budget under ₹23,000.
- Authored a full-body PyBullet physics simulation with real-time inverse kinematics (6 legs × 3 joints), three gait patterns (Wave, Tripod, Ripple) with live switching, and an automated stair-climbing sequence.
- Validated servo selection by modeling MG996R specs (11.0 kg-cm stall torque @ 6V) against computed per-joint torque demand across a 0–2 kg payload before purchasing hardware.
- Built real-time telemetry dashboards for per-leg torque and foot height; audited the model and corrected per-leg load distribution and gait timing so Wave/Ripple keep a single leg airborne.
- Designed the electronics: 18 MG996R servos on a PCA9685 PWM driver, BNO055 9-DOF IMU for pitch/roll stabilization, VL53L0X time-of-flight sensor for stair-edge detection, and a single LiPo battery with buck-regulated power.
- Developed the Raspberry Pi 4 control software (headless Linux over SSH, 50 Hz control loop) and an operator dashboard for the robot.

SmartIOF: AI-Enabled Embedded Food Spoilage Monitoring System

Sensor Integration & Embedded Systems

Technologies: ESP32, C++, Gas Sensors (MQ-135), DHT22, Glassmorphism Web Interface

- Designed an embedded monitoring system tracking food storage conditions in real time.
- Integrated temperature, humidity, and VOC (gas) sensors for spoilage detection with non-blocking timing control.
- Implemented responsive local web dashboard displaying live threat levels (1–5 scale).

Group RC Aircraft Design & Fabrication

Aerodynamic Design & Assembly (Team Project)

Technologies: Airfoil Selection, Aerodynamics, RC Systems, Servo Motors, Brushless Motors

- Collaborated in a team to design and assemble a functional remote-controlled aircraft.
- Applied aerodynamic principles to select and construct custom wing airfoils for optimized lift-to-drag performance.
- Integrated brushless motors, ESCs, servo motors, and multi-channel RC receiver with center-of-gravity balancing.

Peaucellier Linkage Mechanism

Mechanical Design & Kinematics (Group of 5)

Technologies: Kinematics, Mechanical Design, Motion Linkages, Prototyping

- Designed and built a physical mechanism converting rotary motion into precise straight-line motion verified through iterative testing.

EDUCATION

Bachelor of Engineering in Mechatronics Engineering
Mangalore Institute of Technology & Engineering (MITE)
2023 – Present
CGPA: 6.98

Senior Secondary (12th Grade)
St. Aloysius Pre-University College, Mangalore
2021 – 2022
Score: 68.8%

Secondary School (SSLC / 10th Grade)
St. Aloysius High School, Mangalore
2019 – 2020
Score: 73.9%

CERTIFICATIONS & WORKSHOPS

- **NX Essentials for NX Designer** — Certificate of Merit, MITE-Siemens Centre of Excellence (Aug–Dec 2025)
- **Robcad Basics** — Certificate of Merit, MITE-Siemens Centre of Excellence (Jan–May 2026)
- **Image Processing: Fundamentals to Intelligent Applications** — Workshop, MITE Dept. of Mechatronics (Sept 2026)
- **Electric Vehicle Development using Model-Based Design** — CoreEL Technologies (2024)
- **Prototype Development of Control Circuits for Electric Drive Systems** — CoreEL Technologies (2025)
- **Automation Explorer SKILL-A-THON** — UiPath Academic Alliance (2024)
- **AI & R Programming Series** — Infosys Springboard
- **Salesforce Agentblazer Champion** — Salesforce Trailhead (2025)

ACHIEVEMENTS

- **e-Yantra Innovation Challenge (2025–26):** Reached Round 3 in national competition by IIT Bombay.
- **Hackathons:** Participated in SAP Hackfest & Smart India Hackathon (2024).